

Terrain-Interaction Control of a Hexapod Robot: a Measured Comparison of Three Controllers on a Shared Simulation Plant

[Student 1]

[Student 2]

Field and Service Robotics — A.Y. 2025/2026

Università degli Studi di Napoli Federico II

October 4, 2026

Abstract

We compare three controllers for the PhantomX AX Metal Hexapod Mark II on a single high-fidelity Simscape Multibody plant: an open-loop kinematic tripod controller (C1), the torque-based terrain-search feedback of Arrigoni et al. [1] (C2), and an attitude loop added on top of it (C3). Each controller adds exactly one mechanism to the previous one, so that every adjacent comparison isolates a single cause.

The headline outcome of the project was not in the original plan. Roughly half of the effort went into *finding and correcting two defects of our own test bench*, both capable of inverting the conclusions and neither of which announced itself: the robot walked, the animation was plausible and the metrics came out. We report them in full, together with the measurement discipline we adopted afterwards — a measured noise floor, a readability threshold fixed before looking at the data, and written retractions — which we consider the most transferable contribution of this work.

With that discipline in place, the measured result is that terrain search *costs* body attitude where there is nothing to search for and *repays* it on discontinuities, while the attitude loop gives that attitude back and pays for it in energy. Neither mechanism changes the response to a lateral impulse, and none of the three completes the seven-obstacle course.

Contents

1	Introduction	3
1.1	The platform and the reference work	3
1.2	Why we selected this project	3
1.3	What this report claims	3
2	Experimental setup	3
2.1	The plant	3
2.2	The three controllers	4
2.3	Gait parameters and where they come from	5
2.4	Task battery	5
3	Methodology: when is a difference a result?	5
3.1	A measured noise floor, and a readability threshold	6
3.2	The criterion is written before the run	6
3.3	Retractions stay written	6

4	Two defects of the test bench	6
4.1	The URDF inertias are wrong by a factor of about one thousand	7
4.2	Correcting the robot broke the contact detector	7
4.3	Re-tuning the contact threshold	8
4.4	A slew-rate limiter, and why it belongs in the plant	8
5	Results	8
5.1	Flat ground and curved path: terrain search costs attitude	8
5.2	Discontinuities: terrain search repays	10
5.3	The ramp: what the attitude loop is for	10
5.4	Obstacle course: the comparison does not discriminate	12
5.5	Lateral impulse: a gap common to all three	12
5.6	The stability envelope in speed	12
6	Feasibility on the real actuators	12
7	Discussion	14
7.1	The two mechanisms do different things, and only one of them pays	14
7.2	A finding about the composition itself: stance detection degrades	15
7.3	What the task battery could not settle	16
8	Limitations	16
9	Contributions by team member	17
10	Relation to other coursework and novelty	17
11	Conclusions	18

1 Introduction

1.1 The platform and the reference work

The PhantomX AX Metal Hexapod Mark II is an inexpensive, open-hardware six-legged robot with eighteen Dynamixel AX-12A smart servos, three per leg (coxa, femur, tibia). In its standard configuration it has *no* force sensors at the feet and no IMU: the only feedback available from the hardware is the position and load estimate reported by each servo.

Arrigoni et al. [1] exploit precisely that constraint. Their contribution is a feedback architecture that detects foot-ground contact *from motor torque alone*, by comparing the measured joint torque against the torque predicted by an inverse-dynamics model of the robot, and uses the detection to adapt the commanded foot height to terrain the trajectory generator does not know about. The appeal of the idea is that it buys terrain adaptation without adding a single sensor.

1.2 Why we selected this project

Three reasons made this an attractive final project rather than a reimplementing exercise.

The claim is falsifiable, and the paper does not quantify its price.

The reference work demonstrates that terrain search improves behaviour on unmodelled terrain. It does not report what the mechanism costs where there is no terrain to discover. A controlled comparison on identical ground, with a declared readability criterion, can answer that — and the answer turned out to be a real and systematic cost (Section 5).

The contact detector is a model-based observer in disguise.

The flag that drives the whole architecture is $\text{OR}_j \left(|\tau_j^{\text{meas}} - \tau_j^{\text{exp}}| > \sigma_j \right)$, where τ^{exp} comes from a rigid-body-tree inverse-dynamics model. Its quality therefore depends on the *accuracy of a dynamic model*, which connects directly to the modelling and state-estimation material of the course. We found out the hard way how directly (Section 4.2).

A hexapod on a shared plant makes the ablation clean.

Because all three controllers drive the same multibody model, the same ground, the same contact law and the same gait generator, each adjacent comparison differs by one mechanism only. That is rarely available when comparing published controllers against each other.

1.3 What this report claims

1. Two defects of the simulation bench, found, measured, corrected and documented, each of which had already produced published-looking numbers that were wrong (Section 4).
2. A measurement protocol that makes a difference between two controllers *declarable or not declarable*, rather than merely observed (Section 3).
3. A three-way comparison over seven tasks with that protocol applied (Section 5).
4. A feasibility check of the commanded torques against the real servo datasheet (Section 6).

2 Experimental setup

2.1 The plant

The robot is imported from the public URDF of the PhantomX description package into Simscape Multibody: 25 rigid bodies (trunk plus 24 leg links), 18 revolute joints, and six spherical contact elements at the feet. Joints are commanded in *position*; the resulting joint torque is therefore a reaction, not an input — a point that matters for Section 6. Foot-ground interaction uses a penalty contact law with Coulomb friction, tuned so that the static penetration (4.3 mm in

tripod stance) sits well above the transition region (2.0 mm), where the force law would otherwise be ill-defined.



Figure 1: The Simscape Multibody plant: body, six three-degree-of-freedom legs, spherical contact elements and the terrain props used by the task battery.

2.2 The three controllers

Table 1: The three controllers. Each adds one mechanism to the previous one.

	Plant	Mechanism
C1	<code>sim_zero</code>	Open-loop kinematic tripod. Foot trajectories generated from the gait parameters, inverse kinematics per leg, joints commanded in position. No feedback of any kind.
C2	<code>sim_zero</code>	C1 plus <i>terrain search</i> : while a leg is in stance and the contact flag is low, the commanded foot height keeps descending until the torque residual exceeds the threshold.
C3	<code>sim_attitude</code>	C2 plus an <i>attitude loop</i> : two discrete PI regulators on body pitch and roll, whose outputs are distributed over the six feet as an additive height offset.

The attitude loop acts *in series*, *after* the terrain search: it reads the already-corrected foot height and adds a term. We verified this by reading the block connectivity inside the model file, not from documentation:

$$\text{terrain_search} \rightarrow z_i \rightarrow \Sigma(+\Delta z_i) \rightarrow \text{Saturation} \rightarrow \text{Rate Limiter} \rightarrow \text{leg}$$

with

$$\Delta z = \mathbf{x} u_\theta + \mathbf{y} u_\phi, \quad (1)$$

where u_θ and u_ϕ are the pitch and roll regulator outputs and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^6$ are the lever arms of the six feet. Pitch uses $P = 10$, $I = 1$; roll uses $P = 5$, $I = 0.5$; both are discrete at 1 ms with zero reference. Because C3 contains C2, the comparison $C2 \rightarrow C3$ isolates the attitude loop, and $C1 \rightarrow C2$ isolates terrain search.

2.3 Gait parameters and where they come from

The reference work delegates gait generation to NUKE and states explicitly that it will not discuss it. The gait parameters therefore had to be chosen. They are not arbitrary: three of five are fixed by a measured constraint, and the remaining degree of freedom is a single one, because

$$T = \frac{S}{\beta_{\text{stance}} v_{\text{nom}}}. \quad (2)$$

Table 2: Gait parameters and their provenance.

Parameter	Value	Origin
phase vector	—	Definition of the alternating tripod: groups {FL, MR, RL} and {FR, ML, RR}. Not a tunable.
β_{stance}	0.50	The only duty factor giving continuous support on <i>exactly</i> three legs. Below it, instants with fewer than three feet exist (outside static stability); above it, six-foot windows appear in which 18 position constraints act on 6 degrees of freedom.
S	0.06 m	Bounded by leg reach. At mid-stride the leg uses 80.9% of $\ell_f + \ell_t = 0.219$ m; the internal assertion trips at 85%, i.e. at $S = 0.095$ m. The 85% bound is itself measured: at 89% the robot walked backwards.
H	0.05 m	1.5× the measured obstacle height (32.8 mm, read from the stance-foot elevation in the single-obstacle run, not from the mesh).
T	1.00 s	Dependent variable, via (2). Gives $v_{\text{nom}} = 0.12$ m/s, verified at 83% of the measured stability envelope (Section 5.6).

The resulting Froude number is $v^2/(gh) = 0.12^2/(9.81 \times 0.154) = 0.0095$: inertial forces are one percent of gravity. The march is firmly quasi-static, which is what legitimises the *static* stability assumption the tripod gait is built on.

2.4 Task battery

Table 3: The seven tasks.

ID	Name	What it exercises
T2	flat, speed sweep	Baseline, and the stability envelope in speed.
T3	curved path	Yaw command 0.1 rad/s.
T4	8° ramp	Continuous slope: body attitude versus terrain.
T4D	8°/8° bump	Up-then-down slope, two angular interfaces.
T5	single obstacle	One unmodelled step, 32.8 mm.
T6	seven obstacles	Repeated unmodelled discontinuities, half-width so that left and right sides are raised alternately.
T7	lateral impulse	Momentum step from 1% to 1600% of nominal.

3 Methodology: when is a difference a result?

Three rules, each adopted after having violated it at least once. We regard them as the most transferable part of this work, because they are what separated the results in Section 5 from the numbers we believed before.



Figure 2: The terrain configurations of the task battery.

3.1 A measured noise floor, and a readability threshold

Every run is repeated three times with the stance height z_0 displaced by ± 0.2 mm — a quantity that has nothing to do with the controller and everything to do with where the ground happens to be. The resulting excursion of each metric is the *noise floor* of that metric, on that task, for that controller.

A difference between two controllers is declared only if it is at least **three times** the worse of the two noise floors.

This is not a formality. Applying it removed several differences that looked like results; applying it *badly* — with a floor measured on a different controller — made us believe for a day that nothing at all was declarable. The floor is measured on four tasks (flat, curve, bump, obstacle) for all three controllers, each against itself.

Throughout Section 5, a figure in parentheses such as (16.3×) is the ratio of the observed difference to that noise floor. Entries marked n.c. (not conclusive) fall below 3×; entries marked n.m. have no measured floor on that task and are reported without a verdict.

3.2 The criterion is written before the run

Every diagnostic script in the repository carries, in its header, the question it asks, the threshold it will apply, and the possible conclusions — all written before the data exist. This matters because the alternative is available and tempting: on this project a readability threshold calibrated after seeing the data would have moved the stability limit from 1.0× to 1.2× depending on which multiple we chose.

3.3 Retractions stay written

Every claim we withdrew is marked [RETRACTED <date>] in place, with the reason. We withdrew at least five during the project. On this work a wrong claim survived three days precisely because it was plausible, and the discipline exists to make that visible rather than tidy.

4 Two defects of the test bench

The largest single block of work in this project was not the comparison. It was discovering that the bench we were measuring with was wrong in two places, and that the second defect was

caused by fixing the first.

4.1 The URDF inertias are wrong by a factor of about one thousand

The plant takes masses and inertias from a public URDF. The masses are correct; the inertia tensors are not. Two independent arguments, neither of which depends on trusting any particular source for the file:

Radius of gyration.

From $I = mr^2$, the declared trunk inertia places the trunk’s mass at $r = 1.79$ m from its own centre. The trunk is 25 cm across. Every link gives $r = 0.46$ m; the links are 2–12 cm long. Read in $\text{kg} \cdot \text{m}^2$, as the URDF standard requires, those numbers describe an object that cannot exist.

All 24 links carry identical inertia,

to the same decimal. Coxa, femur and tibia differ in shape and length. It is a placeholder, not a measurement.

Dividing by 10^3 brings everything into the physically plausible range, which is the signature of a CAD export performed with masses in grams. The effect on conclusions, not merely on values, is large: with the URDF inertias the pitch of C1 on rough ground reached 45° , against 15° with corrected ones, and the cost ratio between C2 and C1 moved from 1.35 to 0.68 — it *changed sign*.

4.2 Correcting the robot broke the contact detector

The contact flag of C2 is not a threshold on measured torque. The model computes

$$\text{contact} = \bigvee_{j \in \{c, f, t\}} \left(\left| \tau_j^{\text{meas}} - \tau_j^{\text{exp}} \right| > \sigma_j \right), \quad (3)$$

and τ^{exp} is produced by an inverse-dynamics block carrying a `rigidBodyTree` imported *from the same URDF*.

We corrected the inertias of the simulated robot only. From that moment the robot and the model predicting its torque were two different robots, with inertias differing by three orders of magnitude. The residual $|\tau^{\text{meas}} - \tau^{\text{exp}}|$ became large *always*, with or without contact, and the flag stayed high over 95% of the swing phase. With the flag stuck at one, the terrain search never enters the branch that grows the foot extension, and the command degenerates into a constant.

For three days, C2 did not search the terrain. Nothing signalled it: the robot walked, the animation was plausible, the metrics came out. Every C2 row produced in that window described a broken controller.

How we measured it. A single ten-second run, in which the log contains simultaneously the measured torque, the expected torque and the *normal force at the foot*. The flag can then be recomputed offline for any threshold and compared against the force, which is ground truth.

The fix. A routine rewrites the mass and inertia of each body of the estimator’s rigid-body tree with the same corrected values applied to the plant, forces the masked subsystem to re-evaluate, and verifies in three successive ways — write, fallback, read-back — stopping with an error if any of them fails. A silent failure here would restore exactly the problem it closes. It is called immediately after the inertia correction in every task script, in memory, never saving the model.

Table 4: Contact flag against measured foot force, at unchanged threshold. The second row is the diagnosis: a flag that is always high is wrong exactly as often as ours was, so ours carried no information.

	estimator misaligned	aligned
flag error during stance	7.0%	0.6%
error of an <i>always-high</i> flag	7.0%	3.2%
does the flag carry information?	no	yes
swings with no reset of the search	20/60	2/60

4.3 Re-tuning the contact threshold

With the estimator aligned the flag works, but at the original threshold $[0.04, 0.1, 0.1]$ it still stays high over 41.9% of swing, and 39.7% of that is the coxa alone: its p_{90} in swing (0.0421) overlaps its p_{10} in stance (0.0469). This has a physical explanation — the coxa rotates about the vertical axis while contact is vertical, so contact transmits almost no torque to it, while in swing it carries the whole lateral acceleration of the leg.

The adopted triple is $[0.385, 0.136, 0.105]$, which sets the coxa threshold above its own stance median, i.e. switches it off. **This is a change of rule, not a re-tuning:** the flag becomes an OR over two joints. It was decided on the obstacle course, where the outcome is binary and not subject to measurement noise, and it must be declared as our choice: it is no longer the value present in the reference implementation.

4.4 A slew-rate limiter, and why it belongs in the plant

The attitude model originally carried a rate limiter of ± 0.3 m/s on each foot height command, which the open-loop plant did not have. That is an asymmetry: the limiter constrains *every* height command, including the one produced by terrain search, so a C2–C3 comparison would have confounded the attitude loop with its own limiter.

We therefore placed the same limiter, at ± 0.4 m/s, in both plants. The value is not free: the nominal gait peaks at 0.375 m/s of vertical foot speed, so at ± 0.4 the nominal march is untouched and the limiter acts only on fast commands — terrain search and attitude corrections. We verified that the C1 rows and the C1 noise floor are byte-identical with and without the limiter on five of the seven tasks and on the nominal cell of the sixth.

This turned out to matter more than expected: the limiter removed the lateral drift of C2 on the obstacle course (from 0.296 to 0.104 m) and an anomalous noise sensitivity of the energy of C3. An implementation detail, not a controller, with effects larger than some of the differences between controllers.

5 Results

All rows below come from a single campaign run after the corrections of Section 4, with the slew-rate limiter present on all three plants: seven tasks $\times$ three controllers, plus a noise floor of four tasks $\times$ three runs $\times$ three controllers. Verdicts in bold pass the $3\times$ rule of Section 3.1.

5.1 Flat ground and curved path: terrain search costs attitude

On ground that is exactly what the trajectory generator believes it to be, terrain search makes the body attitude *worse* by a factor of three to four, and buys nothing: energy and cost of transport are unchanged within noise on flat ground, and improve by two to three percent on the curve. The mechanism keeps working where there is nothing to find, and the body pays for it.

Table 5: Flat ground (T2) and curved path (T3), nominal speed. Ratio to the noise floor in parentheses; n.c. = below $3\times$.

	C1	C2	C3	C1 $\rightarrow$ C2	C2 $\rightarrow$ C3
<i>T2 — flat, nominal speed</i>					
pitch max [deg]	0.514	1.95	0.356	+279% (16.3)	−82% (18.1)
roll max [deg]	0.206	0.280	0.205	+36% (n.c.)	−27% (n.c.)
cost of transport	0.888	0.865	1.11	−3% (n.c.)	+28% (27.7)
energy [J]	18.6	18.2	23.3	−2% (n.c.)	+28% (28.4)
torque RMS [N m]	0.400	0.403	0.372	+1% (n.c.)	−8% (8.0)
peak torque [N m]	1.87	2.02	2.23	+8% (n.c.)	+11% (n.c.)
mean speed [m/s]	0.135	0.136	0.135	+1% (6.9)	−0% (n.c.)
<i>T3 — curve, +0.1 rad/s</i>					
pitch max [deg]	0.579	1.85	0.392	+219% (40.0)	−79% (45.9)
roll max [deg]	0.224	0.351	0.210	+57% (4.6)	−40% (5.1)
cost of transport	0.970	0.951	1.22	−2% (3.6)	+28% (24.3)
energy [J]	30.3	29.5	37.8	−3% (6.4)	+28% (22.5)
torque RMS [N m]	0.398	0.389	0.376	−2% (7.0)	−3% (7.3)
peak torque [N m]	1.91	2.11	2.43	+10% (n.c.)	+15% (n.c.)

[FIGURE PLACEHOLDER]
grafico beccheggio vs tempo, T2, tre curve sovrapposte

Figure 3: Body pitch over two gait cycles on flat ground, for the three controllers. The oscillation of C2 is the terrain search acting on ground it already knows.

5.2 Discontinuities: terrain search repays

Table 6: Bump (T4D) and single obstacle (T5).

	C1	C2	C3	C1 → C2	C2 → C3
<i>T4D — bump, 8°/8°</i>					
pitch max [deg]	8.65	10.4	1.98	+ 21% (19.8)	− 81% (93.8)
roll max [deg]	2.12	1.49	0.611	− 30% (6.4)	− 59% (7.6)
cost of transport	1.12	1.04	1.45	− 7% (5.1)	+ 40% (16.8)
energy [J]	64.1	59.9	83.6	− 7% (3.5)	+ 40% (18.8)
mean speed [m/s]	0.130	0.133	0.131	+ 2% (3.6)	− 1% (3.0)
<i>T5 — single obstacle, 32.8 mm</i>					
pitch max [deg]	7.56	6.41	1.73	− 15% (16.4)	− 73% (19.8)
roll max [deg]	4.49	2.31	1.13	−49% (n.c.)	−51% (n.c.)
cost of transport	1.20	1.08	1.40	− 10% (5.2)	+ 30% (17.1)
energy [J]	48.1	44.3	57.5	− 8% (3.6)	+ 30% (17.2)
torque RMS [N m]	0.408	0.403	0.380	−1% (n.c.)	− 6% (9.9)
peak torque [N m]	4.46	8.33	4.01	+87% (n.c.)	−52% (n.c.)
obstacle cleared	yes	yes	yes	same	same

Here the picture reverses, and this is the result that reproduces the claim of the reference work: on a discontinuity the search improves roll, energy, cost of transport and task completion. One solid exception remains on the bump: **peak pitch is 21% worse**, at a ratio of 19.8. On a continuous slope the search extends the downhill feet and the body follows the gradient — it adapts to the terrain rather than resisting it.

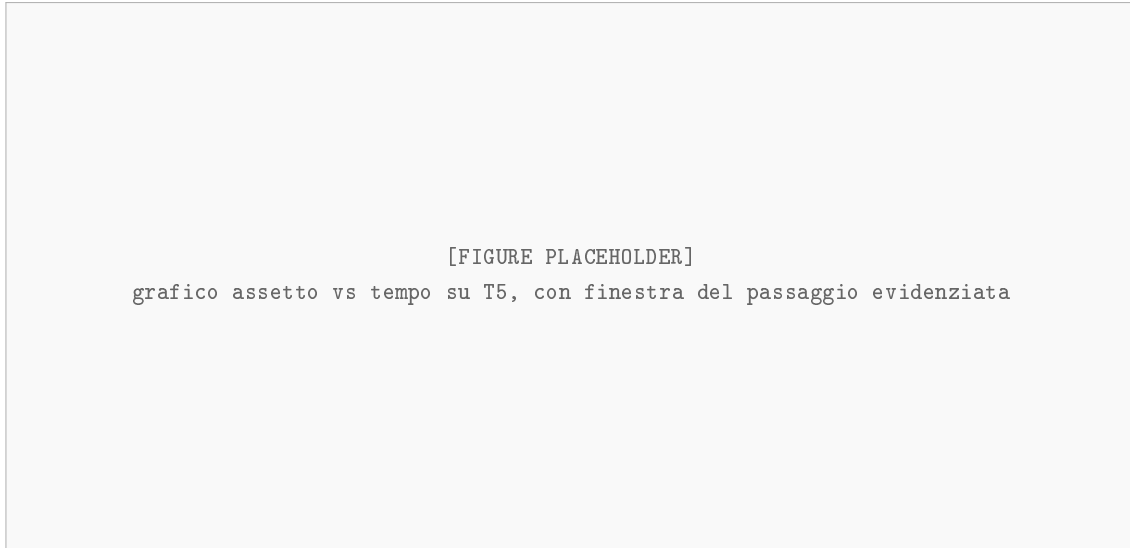


Figure 4: Body attitude during the obstacle crossing (T5). Shaded: the window in which at least one foot is on the obstacle, identified kinematically.

5.3 The ramp: what the attitude loop is for

C1 and C2 align the body with the ramp, to within a few tenths of a degree. C3 holds it 7.54° away from the ramp, i.e. within 0.5° of *horizontal*, which is exactly what a loop with zero attitude reference is built to do — and it is also the behaviour the reference work states as its own objective. It costs 45% in cost of transport.

Table 7: 8° ramp (T4). Noise floor not measured on this task; no verdicts are issued, but the behaviour is qualitative and unambiguous.

	C1	C2	C3
body attitude minus ramp angle [deg]	0.106	0.259	−7.54
peak-to-peak pitch on the ramp [deg]	0.994	1.97	0.839
roll on the ramp [deg]	0.662	0.563	0.314
cost of transport	1.15	1.09	1.58
climb completed	yes	yes	yes



Figure 5: Body pitch while climbing the 8° ramp. C1 and C2 follow the slope; C3 holds the trunk horizontal.

5.4 Obstacle course: the comparison does not discriminate

Table 8: Seven-obstacle course (T6). Only the binary outcome and its validity are reported.

	C1	C2	C3
distance travelled [m]	2.91	3.12	2.99
max lateral drift [m]	0.067	0.104	0.058
course completed, valid	no	no	no

“Completed” counts only if the maximum lateral drift stays below 0.183 m; beyond that the robot may have missed an obstacle entirely rather than stepped over it. The threshold comes from the obstacle geometry, computed independently of any controller.

None of the three completes the course. The fine metrics of this task must not be quoted: the three controllers follow different trajectories and therefore do not meet the same obstacles. On this task the comparison does not separate the controllers, and saying so is the result.

5.5 Lateral impulse: a gap common to all three

Table 9: Residual lateral deviation after an impulsive disturbance (T7).

	C1	C2	C3
residual deviation, 1× impulse [mm]	1.34	6.04	0.164
residual deviation, 16× impulse [mm]	393	400	392
recovers at 16×	no	no	no

At large impulse the three curves coincide to within 2%: none of the three recovers a measurable fraction of the deviation. Contact detection is not the mechanism in play — this is an architectural gap shared by all three, and the clearest motivation for any future controller on this platform.

5.6 The stability envelope in speed

A sweep of six speed factors, judged on body bounce with a threshold declared beforehand, locates the envelope of the open-loop gait.

Table 10: Body bounce versus commanded speed factor, C1. Threshold 10 mm, declared before the sweep.

factor	bounce [mm]	
1.00	3.2	nominal operating point
1.20	3.4	last cell with stable body motion
1.30	9.4	at the threshold
1.40	19.7	fails

6 Feasibility on the real actuators

The campaign runs with ideal actuators: the simulator grants any torque. The AX-12A stalls at 1.5 Nm, and the manufacturer recommends operating at one fifth of stall for stable motion. Since the joints are commanded in position, torque is a reaction rather than an input, so what follows measures what each controller *demand*s, not how it would behave with real motors.



Figure 6: Speed sweep: body bounce and task fraction versus commanded speed factor.

Table 11: Commanded torque against the AX-12A limit. RMS is the worst-loaded joint of each run, as a fraction of stall; “peak” is the maximum torque as a multiple of stall; “over” is the fraction of samples above stall.

task	RMS / stall			peak / stall			samples over		
	C1	C2	C3	C1	C2	C3	C1	C2	C3
T2	46%	44%	39%	1.2	1.3	1.5	0.40%	0.45%	0.38%
T3	46%	42%	39%	1.3	1.4	1.6	0.50%	0.40%	0.43%
T4	44%	44%	45%	1.4	2.3	1.8	0.39%	0.49%	0.48%
T4D	42%	42%	39%	1.7	2.7	2.1	0.45%	0.41%	0.49%
T5	45%	44%	40%	3.0	5.6	2.7	0.61%	0.50%	0.51%
T6	42%	43%	42%	5.0	4.9	2.8	0.91%	0.67%	0.81%
T7	46%	44%	39%	1.2	1.3	1.5	0.41%	0.45%	0.38%

Two regimes, and they must be read separately:

- **The march exceeds the recommended load.** The worst-loaded joint sits between 39% and 46% of stall on every task and for every controller — $2.3\times$ the one fifth of stall the manufacturer indicates for stable motion. It stays below stall, but not within the recommended envelope.
- **The impact peaks leave stall**, up to $5.0\times$ on the obstacle course, but on at most 0.91% of samples: these are isolated events, one per foot strike, lasting tens of milliseconds.

C3 consistently demands the lowest continuous torque (39–45%) and the lowest impact peaks on the tasks with discontinuities, which is consistent with an attitude loop that absorbs part of the impact in body motion rather than in the joints.

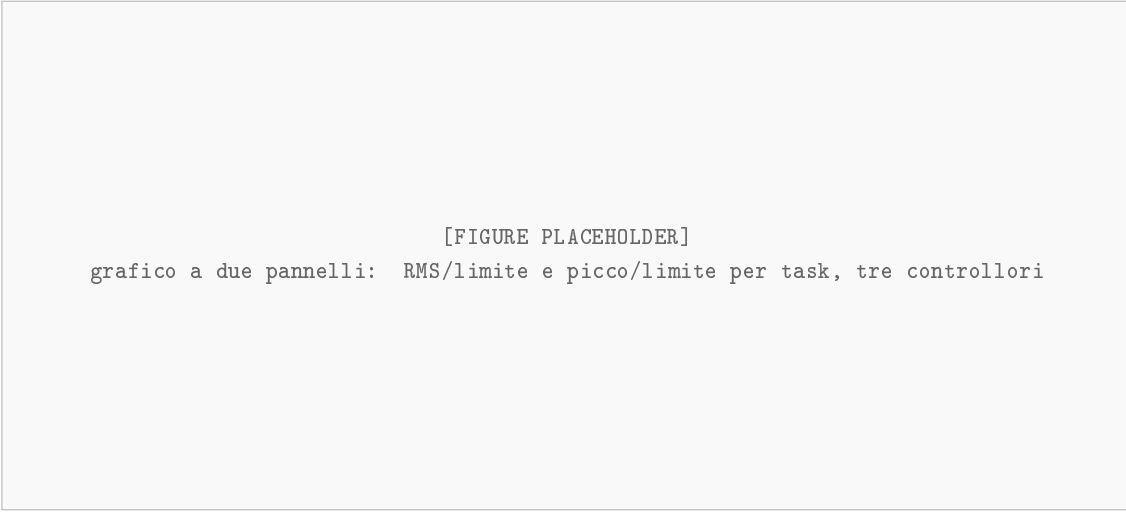


Figure 7: Commanded torque against the AX-12A limit: continuous (RMS of the worst joint) and peak, per task and controller.

7 Discussion

7.1 The two mechanisms do different things, and only one of them pays

Reading Tables 5–11 along the two adjacent comparisons gives a consistent picture across all seven tasks.

Terrain search alone (C1→C2) is not an attitude mechanism. On the two tasks where the ground is exactly what the trajectory generator assumes — flat (T2) and curved (T3) — it makes pitch three to four times worse, by a margin 16 to 40 times the noise floor, and returns nothing measurable in energy. On the tasks with real terrain it does better: roll improves by 30% on the bump and 49% on the obstacle, cost of transport drops 7–10%, and the mean speed rises by 2–3%. But the pitch, which is the degree of freedom the search acts on, still degrades by 21% on the bump.

The reason is visible in the mechanism rather than in the numbers. The search lowers a swinging foot until the torque residual declares contact; the declaration is made per leg, asynchronously, and the resulting height correction is a step. On ground the generator already models correctly the steps are spurious, and three legs stepping at slightly different instants is exactly a pitch excitation. Arrigoni et al. [1] introduce the mechanism for terrain the robot cannot see; our measurement says that on terrain it *can* see, the mechanism is a net cost, and that cost does

not disappear when the terrain becomes irregular — it is simply joined by a benefit on the other axes.

Attitude feedback (C2→C3) is the mechanism that buys attitude, and it buys it with energy. The sign is the same on every task and every magnitude clears the $3\times$ rule by a wide margin: pitch -82% (flat), -79% (curve), -81% (bump), -73% (obstacle), with noise ratios from 18 to 94. On the 8° ramp it is qualitative rather than incremental: C1 and C2 carry the trunk parallel to the ramp (0.11° and 0.26° of body-minus-ramp angle), while C3 holds it within half a degree of *horizontal* (-7.54° of body-minus-ramp on an 8° slope). Those are two different behaviours, not two qualities of the same behaviour, which is why that row carries no percentage verdict.

The price is equally consistent: $+28\%$ energy and cost of transport on flat and curved ground, $+40\%$ on the bump and the ramp, $+30\%$ on the obstacle, all at 17 to 28 times the noise floor. Interestingly the *continuous* torque goes down (-3 to -8% , declarable on five tasks): the extra energy is spent on the additional vertical travel of the feet, not on holding larger joint loads.

The composition is not the sum. C3 is C2 with the attitude loop in series after the terrain search, so the $+279\%$ pitch of the search and the -82% of the attitude loop are measured on the same signal path. Comparing the two ends, C3 ends up *better than C1* on pitch (0.356 vs 0.514 deg on flat), i.e. the attitude loop more than pays back the disturbance the search introduced. Whether a controller consisting of attitude feedback *without* terrain search would do better still is the obvious next ablation, and we did not run it: the plant for it does not exist, and building it would have meant modifying a model that was already frozen for the campaign.

7.2 A finding about the composition itself: stance detection degrades

The clearest architectural result of the project was not planned. When the ramp task (T4) was analysed on C3, the kinematic stance detector (Section 2) — a foot counts as in stance when its world speed stays under 0.05 m/s for at least a quarter of the nominal stance — found essentially no stance at all. Our first reading was that the robot was not walking. It was wrong, and the record of why is worth keeping: the mean body height we quoted was the ramp’s own height, and the force-sensor stance count of 1.00 is meaningless on a ramp, because those sensors only see the floor plane.

The correct reading came from counting, per leg, the fraction of samples under threshold and the longest consecutive run:

	samples under 0.05 m/s	longest consecutive stretch
C2	41–54%	71–113 samples ($\approx$ one full stance)
C3	4–8%	3–12 samples

The feet are not failing to land: they are landing and then being moved again. The attitude loop adds a continuous height correction $\Delta z = x u_\theta + y u_\phi$ to every foot, stance feet included, so a foot in stance is never at rest in the world frame. The amplitude is small — with a median foot speed of 0.14 m/s the motion over one fragmented stance is about 17 mm , contained by the saturation and the rate limiter — so the gait is unaffected. What breaks is the *observability of contact by any rest-based criterion*.

This matters beyond our test bench, because the terrain search underneath is itself a contact-detection mechanism. Composing an attitude loop in series after a contact-triggered mechanism feeds the second one a world in which nothing is ever stationary. The composition works here because the terrain search uses a torque residual rather than foot kinematics, and because the attitude correction is rate-limited; it would not survive a stance detector built on foot velocity. We consider this the most transferable thing the project produced.

7.3 What the task battery could not settle

Three of the seven tasks returned no verdict, and saying so is part of the result.

- **T6 (seven-obstacle course) does not discriminate.** None of the three controllers completes it: 2.91, 3.12, 2.99 m travelled, all three below the validity threshold on lateral drift (0.183 m, derived from the course geometry, not from the controllers). Beyond that drift the robot may have gone *around* obstacle 3 rather than over it, so neither the distance nor any downstream metric can be compared: the three controllers did not meet the same obstacles. This is a change from an earlier campaign, before the slew-rate limiter was present on all three plants, in which C2 appeared to reach 3.71 m against 2.44 m for C1. That headline is superseded and should not be quoted.
- **T7 (lateral impulse) separates only at the small amplitude.** At $1\times$ the residual deviation is 1.34, 6.04, 0.164 mm — a 97% improvement for C3 — but the noise floor was not measured on this task, so no verdict is declared. At $16\times$ all three end up at 392–400 mm and none recovers: the disturbance is simply larger than any of the three can reject.
- **Peak torque never clears the $3\times$ rule,** on any task or any comparison, despite differences as large as +87%. Peak is a single-sample statistic of an impact and its run-to-run scatter is of the same order as the differences between controllers. The RMS column, which is an integral, is declarable on five tasks. We report both and draw conclusions only from the second.

8 Limitations

Ideal actuators. The joints are commanded in position and the simulator supplies whatever torque the command requires. Section 6 therefore measures *demand*, not achievable behaviour. A campaign with the AX-12A’s speed and torque limits in the loop would likely compress the differences, because C3 is the controller that asks for the most additional foot travel per unit time.

The contact sensors see only the floor. The Simscape force sensors are attached to the ground plane, so on the ramp, the bump and the obstacles they report nothing. Every stance-dependent metric on T4, T4D, T5 and T6 comes from the kinematic detector instead — which, as Section 7 shows, is itself degraded on C3. Where the two detectors disagree we report the disagreement rather than picking one.

The attitude lever arms are hardcoded. The x and y used in $\Delta z = x u_\theta + y u_\phi$ are constants in the model, about $1.8\times$ smaller than the nominal foot positions `cfg.pf_nom`, though with correct signs and leg ordering. We chose *not* to correct them: the proportional gains were tuned empirically with those values in the loop, so changing the lever arms without re-tuning would change the loop gain by the same factor. The pair (lever arm, gain) is consistent; only its factorisation is arbitrary.

The inverse-dynamics estimator is not the plant. After the correction of Section 4.2 the residual mismatch is the 1.6% of mass in the feet, which the URDF tree does not carry. The contact threshold σ_j absorbs it, but it was tuned against this bench and would have to be re-tuned on another.

The threshold is self-tuned. σ_j was set from the residual observed during known-free swing on this plant. It is therefore not an independent parameter, and the terrain search cannot be credited with robustness to a threshold it was fitted to.

Quasi-static stability only. The Froude number of the nominal march is $v^2/(gh) = 0.0095$, so the ZMP collapses onto the ground projection of the centre of mass to within the precision of the measurement; the stability analysis assumes a planar contact patch and reports the stance-plane tilt so that the assumption can be checked rather than trusted.

No hardware. Nothing in this report was measured on a physical PhantomX. The platform, the URDF and the servo datasheet are real; the comparison is not validated.

9 Contributions by team member

TODO — to be confirmed and completed by the authors before submission. The split below reflects the division of work as it actually happened; names and any item either of us wants to reclaim must be fixed by hand.

Student 1 — plant and controllers. Construction of the Simscape Multibody plant from the PhantomX URDF; implementation of the open-loop tripod generator (C1); implementation of the terrain-search layer after Arrigoni et al. (C2), including the inverse-dynamics residual and the per-leg contact flag; design, implementation and empirical tuning of the attitude loop (C3) — pitch first, then roll — and of the saturation and rate limiter that bound its authority. Ownership of the Simulink models throughout.

Student 2 — measurement, diagnosis and analysis. Design of the seven-task battery and of the measurement protocol (Section 3), including the noise-floor procedure and the $3\times$ declarability rule; the full measurement campaign and its automation (single-point controller selection, batch execution, result conversion, comparison table generation); identification, quantification and correction of the two test-bench defects of Section 4; the stance-detection diagnosis of Section 7; the quasi-static stability check; the feasibility analysis against the servo datasheet; repository structure and documentation.

Joint. Choice of the three controllers and of the ablation structure; reconstruction and justification of the gait parameters (Section 2); interpretation of the results; this report.

10 Relation to other coursework and novelty

This project was not derived from, and does not extend, a project submitted for another course. The platform model, the three controllers, the task battery, the measurement protocol and all the results reported here were produced for Field and Service Robotics.

Relative to the course material, the project is an application rather than an extension of the theory, and its novelty lies in three places:

1. **A controlled ablation instead of a demonstration.** The usual shape of a project of this kind is: implement a controller, show that it walks. Here the same plant, ground, contact law and gait generator carry three controllers that differ by exactly one mechanism each, so each comparison isolates one mechanism. That design is what makes it possible to say that terrain search costs attitude and attitude feedback costs energy, rather than that one full controller is better than another.
2. **A declarability criterion fixed before the campaign.** Simulation results are reproducible to the last digit if nothing is perturbed, which makes every difference look significant. Perturbing the initial height by $\pm 0.2\text{mm}$ over three runs gives a noise floor, and requiring a difference to exceed $3\times$ the worse of the two floors before it is declared converts a table of numbers into a table of claims. Roughly a third of the differences we measured do not survive that rule, including every peak-torque comparison.
3. **Two silent bench defects, found by measurement rather than by inspection.** Both (Section 4) produced plausible, publishable-looking numbers while being wrong, and neither was visible in the block diagram. The estimator-misalignment one in particular is a direct instance of the course's point about model-based residuals: correcting the plant without correcting the model that predicts it is worse than correcting neither, because the residual saturates and the contact flag degenerates to a constant.

11 Conclusions

Three controllers for a PhantomX hexapod were compared on one plant over seven tasks, under a protocol that declares a difference only when it exceeds three times the measured noise floor.

- The terrain search of [1], taken on its own, is a net cost on terrain the gait generator already models: pitch worsens by 219–279% with no energy return. On irregular terrain it improves roll (−30 to −49%), cost of transport (−7 to −10%) and speed (+2 to +3%), while still worsening pitch.
- The attitude loop is the mechanism that buys attitude, and it does so on every task: pitch −73 to −82% at 18 to 94 times the noise floor, and on the 8° ramp a qualitatively different behaviour (trunk held horizontal rather than parallel to the slope). It costs 28–40% in energy and cost of transport, spent on foot travel rather than on joint load — the continuous torque actually falls.
- Composed in series, the two mechanisms leave the body better than the open-loop baseline on pitch, but they make foot contact unobservable to any rest-based detector: stance stretches collapse from 71–113 samples to 3–12.
- On the real servos, the march itself would run at 39–46% of stall on the worst-loaded joint — 2.3× the manufacturer’s recommended continuous load — with impact peaks above stall on at most 0.91% of samples.

The obvious continuation is the fourth cell of the ablation — attitude feedback without terrain search — which would say whether the search contributes anything at all once the body is stabilised, and a campaign with the servo limits in the loop, which is where the energy cost of C3 would stop being free.

References

- [1] S. Arrigoni, F. Braghin, et al., “Control of a Hexapod Robot Considering Terrain Interaction,” *Robotics*, vol. 13, no. 10, art. 142, 2024.
- [2] ROBOTIS, *Dynamixel AX-12A e-Manual*, ROBOTIS Co., Ltd. <https://emanual.robotis.com/docs/en/dxl/ax/ax-12a/>
- [3] Trossen Robotics, *PhantomX AX Metal Hexapod Mark II — assembly and specifications*.
- [4] The MathWorks, Inc., *Simscape Multibody User’s Guide*, R2025b.